Calibrated Cascade Routing for On-Device Intent Understanding: A Pre-Registered Evaluation on a Laptop CPU

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Abstract

A small language model (SLM) can now run on an ordinary laptop, but sending every request of a personal assistant through it takes close to a second per request on the CPU and more than 3 GiB of memory. We evaluate a calibrated cascade for intent understanding that answers a request at the cheapest tier that is confident enough: keyword rules, then a frozen multilingual sentence encoder with a temperature-scaled linear head, then a 4-bit Gemma-4-E2B model whose confidence comes from constrained label decoding. Every threshold and temperature is fitted on calibration and validation data, and the protocol was pre-registered and frozen before any test item was read. On a six-core laptop CPU, on the CLINC150 test set ($n = 5500$; there the cascade is encoder then SLM, since the rules do not cover its labels), the cascade was more accurate than always using the SLM (Δaccuracy $+0.048$, 95% CI $[+0.038, +0.058]$, non-inferiority margin $\delta = 0.0263$) and faster (Δmean latency -717 ms per request, CI $[-739, -694]$), so the primary hypothesis is supported; temperature scaling lowered selective risk at 80% coverage (Δ -0.0209, CI $[-0.0263, -0.0148]$). The encoder alone, however, was both more accurate and faster than the cascade on CLINC150, and on a small synthetic English–Hinglish set ($n = 252$) the cascade stayed below the SLM (0.968 vs 1.000; inconclusive). An exploratory failure analysis traces 68% of the cascade's CLINC150 errors to out-of-scope requests, whose share rose from 11.8% in validation to 18.2% in test. All results come from one machine and one model per tier; two pre-registered hypotheses (personalization and quantization) are still pending.

Keywords: on-device inference; cascaded inference; calibration; small language models; intent detection; code-mixed Hinglish; privacy.

1 Introduction

A personal assistant for one's own notes and tasks handles text that its user may not want to leave the machine. Running the whole assistant locally removes that exposure, and quantized SLMs of a few billion parameters now run on a laptop CPU through runtimes such as llama.cpp (ggml-org, 2026). The price is cost per request. On the laptop used here, classifying one request with Gemma-4-E2B (Gemma Team, 2026) at 4-bit takes 865.0 ms (median) with 3491 MiB resident, while a frozen sentence encoder answers in a few milliseconds (Table 2). Most requests to such an assistant (*search my notes for budget*, *remind me to call Asha tomorrow*) are simple, so a natural design sends each request to the cheapest component that is confident enough and escalates only the rest. This is a model cascade (Varshney and Baral, 2022; Chen et al., 2024).

Cascades and routers are well studied, but rarely in this setting. In the closest prior work (Table 1), cost is measured as FLOPs or parameter counts rather than as wall-clock time and memory on a device (Li et al., 2021; Lu et al., 2024; Zhang et al., 2026). "On-device" routing has been studied on data-centre GPUs only (Chuang et al., 2025), and the routing signal for intent detection is uncalibrated (Arora et al., 2024). Prior calibrated cascades also chain models of one kind, all encoders or all language models, that mostly score one shared label set. None of them

Table 1: Closest prior work, compared on the columns of our search protocol. "Cost on hardware" means wall-clock time, memory or energy measured on the target device, as opposed to FLOPs, parameter counts or API latency. Entries marked with † were checked against the full text; the others against the abstract.

Work	What is cascaded	Calibrated	Cost on hardware	On device	Rule tier	Code-mixed
C3 (Lu et al., 2024)†	encoders or LLMs of increasing size	yes	no (FLOPs)	no	no	no
CascadeBERT (Li et al., 2021)†	BERTs of increasing depth	yes	no (FLOPs)	no	no	no
CAUC (Zhang et al., 2026)†	2–7 models, same labels	yes	no (parameters, GFLOPs)	no	no	no
Arora et al. (2024)†	encoder → cloud LLM	no	API latency	no	no	no
Chuang et al. (2025)†	SLM → LLM	UQ scores	no (A100s)	no	no	no
Chaganti (2026)†	NER encoder → larger	yes	latency	yes	no	no
Sadhu et al. (2026)†	category router → solver / SLM	n/a	latency, energy	yes	yes	no
This work	rules → encoder → SLM	yes	latency, RAM	yes	yes	Hinglish

mixes a rule tier that may decline to answer, a classifier head and a generative model. Finally, routing surveys warn that cost-optimized routing may serve under-represented languages and non-standard writing worse (Moslem and Kelleher, 2026), but code-mixed input such as Hinglish is untested in this line of work.

We therefore *evaluate*, rather than propose, a calibrated cascade in this setting. Following the novelty discipline of our protocol, we make no claim that the combination is new. Each component has close prior work (Section 2). The questions are: **RQ1**: at a validation-chosen operating point, is the calibrated cascade as accurate as always using the SLM, and faster? **RQ2**: do calibrated confidences lower selective risk? **RQ3**: what does each tier contribute? **RQ5**: how much does a small trainable head forget when new intents arrive, and does replay help? **RQ7**: does the cascade serve Hinglish requests as well as English ones? (RQ4, on personalization, and RQ6, on quantization, are pre-registered but have not run; Section 13.)

Contributions.

1. A pre-registered evaluation of a cascade over three kinds of tier (rules, an encoder head, a generative SLM with constrained label decoding) on a commodity laptop CPU. It reports accuracy together with measured latency and peak memory against baselines that receive the same tuning: the SLM alone, the encoder alone, and cost-matched random escalation.

2. A replication of calibrated thresholding (Lu et al., 2024; Zhang et al., 2026) in this setting: the effect of temperature scaling on selective risk, measured on a public and a synthetic dataset, including the result that did not replicate.

3. An exploratory failure analysis that names two mechanisms behind the cascade's errors: a shift in the out-of-scope rate between validation and test, and a gap in the multi-intent signal.

4. A reproducibility package: frozen configurations, hashed models and data, and a one-command check that regenerates every analysis byte for byte.

The working assistant built along the way (Section 5) is an engineering artifact, not a research claim.

2 Related Work

Model cascades and early exit. Early-exit networks attach classifiers to intermediate layers and stop when a calibrated confidence is high enough (Schwartz et al., 2020; Xin et al., 2020; Zhou et al., 2020). Cascades of complete models of increasing size follow the same idea across models (Li et al., 2021; Khalili et al., 2022; Varshney and Baral, 2022), and Rotem et al. (2023) compare the two designs. C3 (Lu et al., 2024) calibrates each model of a cascade of multilingual

encoders, or of Llama-2 chat models, before thresholding, which is the closest design to our Tier 3 decision, but it measures speed-ups in FLOPs.

LLM routing and cascades. FrugalGPT (Chen et al., 2024), RouteLLM (Ong et al., 2025), AutoMix (Aggarwal et al., 2024) and token-level deferral rules (Gupta et al., 2024) route between language models of different cost, with benchmarks such as RouterBench (Hu et al., 2024). Ramírez et al. (2024) defer on the small model's margin alone, and Dekoninck et al. (2025) unify routing and cascading in one framework. CAUC (Zhang et al., 2026) calibrates every model so that one threshold can be shared across the cascade (a budgeted variant picks it on validation data), and proves that calibration bounds the accuracy of accepted answers; our RQ2 replicates this idea rather than proposing it. Two cautions shaped our protocol. Wang et al. (2026) find that the commercial routers they study do not beat random routing between two well-chosen models at matched cost, so we include a cost-matched random baseline. Jitkrittum et al. (2023) show that confidence-based deferral can fail under distribution shift, among other settings, which motivates the language-stratified analysis (RQ7). A recent survey (Moslem and Kelleher, 2026) covers the field.

Calibration and selective prediction. Temperature scaling (Guo et al., 2017) fits one scalar per model on held-out data. Selective classification (Geifman and El-Yaniv, 2017) evaluates a model that may abstain by its risk at a given coverage, which is the view we take of every system.

On-device inference. Chuang et al. (2025) study SLM-to-LLM routing for on-device settings, Kag et al. (2023) train an edge classifier that defers hard cases to the cloud, and Chaganti (2026) cascade NER encoders on an Apple M4 Pro machine with a cost-matched random baseline. Sadhu et al. (2026) route by category, before inference, to symbolic solvers or a CPU-quantized SLM on a Raspberry Pi and measure energy; their router escalates only when a solver fails, never on low confidence. Islam et al. (2025) measure SLMs on edge boards, and Lim et al. (2026) route requests by intent to prompt-based or lightweight reasoning components.

Intent detection, out-of-scope requests and code-mixing. CLINC150 (Larson et al., 2019) added out-of-scope requests to intent benchmarks, BANKING77 (Casanueva et al., 2020) fine-grained intents, and Zhang et al. (2022) show that in-domain out-of-scope requests are much harder to detect than out-of-domain ones. Sentence encoders such as Sentence-BERT (Reimers and Gurevych, 2019) and multilingual E5 (Wang et al., 2024) give strong frozen features. Arora et al. (2024) route from a SetFit encoder to a cloud LLM on uncertainty, and Rodrigues and Vas (2026) find classifiers cheap where labels exist and LLMs stronger on out-of-scope requests. Code-switching degrades multilingual NLU (Birshert and Artemova, 2022). Intent detection on code-mixed input (Jayarao and Srivastava, 2018), code-switched (Spanish–English) task-oriented parsing (Einolghozati et al., 2021) and LLMs on colloquial code-mixed Hinglish queries (Dey et al., 2026) have been studied, but not inside a cost-aware cascade.

Continual learning. Elastic weight consolidation (Kirkpatrick et al., 2017) penalizes changes to important weights. Replay of stored examples is a strong baseline, and we measure forgetting and backward transfer as defined by Chaudhry et al. (2018) and Lopez-Paz and Ranzato (2017).

3 Problem Formulation

Each request is a text x with a gold label $y \in \mathcal{Y} \cup \{\perp\}$, where $\mathcal{Y}$ is the set of supported intents and $\perp$ marks an out-of-scope (OOS) request; the Kestrel data also has *compound* requests that carry two or three intents. A system returns either a label $\hat{y}$ (an intent in $\mathcal{Y}$, or an explicit OOS label where the label set has one) or an abstention. Scoring follows the pre-registration: an

in-scope item is correct if and only if $\hat{y} = y$, and *abstention counts as an error*; an OOS item is correct if the system abstains or predicts the OOS label; a compound item is correct if it is not accepted as a single intent below Tier 4.

The cascade has an ordered set of tiers $K \subseteq \{1, 3, 4\}$ (Tier 2, a distilled micro-transformer, was not built). Tier k returns a prediction $\hat{y}_k(x)$ with a calibrated confidence $p_k(x) \in [0, 1]$, or no opinion. With per-tier thresholds $\tau = (\tau_k)_{k \in K}$, the cascade accepts at the first tier with $p_k(x) \geq \tau_k$ and abstains if none does. Let c_k be the measured cost of tier k and $\text{reach}_k(\tau)$ the share of requests that consult it. Because every consulted tier is paid for,

$$\mathbb{E}[\text{cost}](\tau) = \sum_{k \in K} \text{reach}_k(\tau)\, c_k, \qquad \text{reach}_{k_1} = 1, \quad \text{reach}_{k_{j+1}} = \text{reach}_{k_j} \cdot P(p_{k_j} < \tau_{k_j}). \tag{1}$$

The operating point minimizes expected cost subject to an accuracy floor set by the tuned single-model baselines, on the validation set V:

$$\tau^\star = \arg\min_{\tau} \ \mathbb{E}_V[\text{cost}](\tau) \quad \text{s.t.} \quad \text{acc}_V(\tau) \ \geq \ \max\big(\text{acc}_V(\text{R0}),\, \text{acc}_V(\text{R1})\big), \tag{2}$$

with ties broken towards higher accuracy. The floor has no tolerance: the non-inferiority margin δ is used only in the test, because a point chosen at the floor minus δ would sit on the margin's edge.

4 Proposed Method

Tiers. *Tier 1* is a hand-written keyword rule set for the Kestrel intents; its confidence is its precision on the calibration split, and it has no opinion on requests no rule matches. *Tier 3* embeds the normalized request with multilingual-e5-small (Wang et al., 2024) (Q8_0 GGUF, frozen) and applies a softmax-regression head (L-BFGS); $p_3 = \max \text{softmax}(z/T_3)$ with T_3 fitted by negative log-likelihood on the calibration split. *Tier 4* prompts Gemma-4-E2B-it (Q4_0) with a fixed template and few-shot examples, and decodes under a grammar that allows only label strings; its confidence is the product, over the constrained decoding steps, of $\text{softmax}(\ell/T_4)$ at the chosen token, where ℓ are the logits of the allowed tokens and T_4 minimizes binary cross-entropy against correctness on the calibration split. For CLINC150 the label set includes an explicit `oos` answer, and the cascade is Tier 3 $\rightarrow$ Tier 4, since the Kestrel rules cannot answer with CLINC150 labels.

Multi-intent signal. A request that contains a conjunction (*and*, *then*, *aur*, *phir*, *fir*) goes straight to Tier 4, which can plan several steps; lower tiers are not consulted. Its false-positive rate on single-intent requests was 4.0% on CLINC150 and 0.0% on Kestrel-Intents-v0.

Threshold selection. Per-tier thresholds are found by grid search over the validation set, using the tier outputs logged once per item and the tier costs measured on the development runs, and Eq. (2) selects the operating point (Figure 1). The whole accuracy–cost frontier on validation is kept and reported.

5 System Architecture

Kestrel is a Rust workspace (Figure 2): `kestrel-core` (types, normalizer), `kestrel-nlu` (Tiers 1 and 3), `kestrel-inference` (Tier 4 behind a `Generator` trait, llama.cpp through `llama-cpp-2`), `kestrel-router` (thresholds, latency budget, traces), `kestrel-tools` (an allow-listed tool set, a path sandbox, plan validation and execution), `kestrel-memory` (one SQLite file for tasks, preferences and memories), `kestrel-learning` (corrections turned into memory proposals that the user confirms) and `kestrel-bench` (timing, peak RSS). The command `kestrel ask --notes`

Route(x) *(deployed; Rust, `kestrel-router`)*

if x contains a conjunction **then** consult only Tier 4
best $\leftarrow$ none
for tier k in K, cheapest first:
  $(\hat{y}, p) \leftarrow$ tier k on x (on error or no opinion: **continue**)
  if $p \geq \tau_k$ **then return** accept$(\hat{y}, p, k)$
  best $\leftarrow$ the more confident of *best* and $(\hat{y}, p, k)$
return abstain(*best*)

Tune *(offline; Python, `experiments/router/policies.py`)*

1. Fit T_3 (NLL) and T_4 (BCE) on the calibration split; Tier-1 confidence = rule precision on it.
2. Tune R0's and R1's single abstention thresholds on validation for maximum accuracy.
3. Grid-search τ on validation; keep the frontier; choose $\tau^\star$ by Eq. (2).
4. Freeze configurations and hashes (git tag) before any test item is read.

Figure 1: The routing rule and how its parameters are fitted. Calibration, thresholds and the operating point use only the calibration and validation splits.

`DIR` runs the assistant on a folder of notes: an accepted intent is planned, the plan is validated, and tools run inside the sandbox; deleting a task asks for confirmation. The runtime has no network crate, no URL and no shell; Section 14 reports the egress checks. The evaluation harness drives the same components.

6 Experimental Setup

Hardware and software. One laptop: AMD Ryzen 5 5500U (6 cores, 12 threads), 14.95 GiB RAM, Fedora 44 (Linux 7.2.6), Rust 1.99.0, Python 3.14.7, llama.cpp commit `26394b4e` through `llama-cpp-2` 0.1.157. Models are loaded without memory mapping, so peak RSS (`VmHWM`) is the real footprint.

Datasets. *CLINC150* (Larson et al., 2019) (full version): train 15 000, calibration 1 500, validation 1 700 (with 200 OOS) and the official test set, $n = 5500$ with 1 000 OOS requests. The 100 official OOS training items join validation, and the 14 test items whose text also occurs in another split are kept, with a sensitivity analysis that drops them. *Kestrel-Intents-v0* is a synthetic set of assistant requests in English and Hinglish (romanized Hindi–English), drafted with an AI assistant from templates and reviewed by the author: train 604, calibration 100, validation 128, test $n = 252$ (126 per language, 44 template families). It is split by template family, its test OOS requests come from unseen domains, and its compound requests occur only in validation and test. It is labelled synthetic in every table.

Pre-registration. Hypotheses, systems, scoring, the measurement protocol, invalidation rules and the analysis were written in a pre-registration and frozen (git tag `exp-E1-frozen`) before any test item was read. One later clarification, which fixed the datasets of three secondary hypotheses, is disclosed (Section 12). The margin $\delta = 0.0263$ is twice the seed-to-seed standard deviation of the tuned SLM's accuracy in a five-seed pilot on validation data; it exceeds the provisional smallest effect of interest, 0.02.

Measurement protocol. AC power, the `performance` governor, turbo boost off, six threads pinned one per physical core, background services stopped, and at least five minutes of cooldown

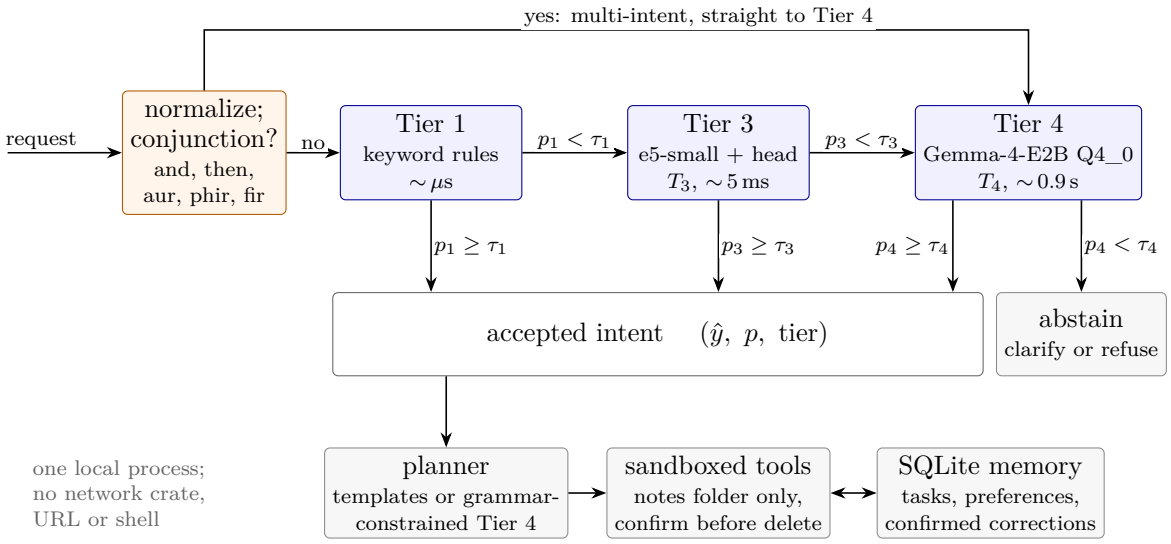


Figure 2: The Kestrel cascade (diagram D1). A request ends at the first tier whose calibrated confidence reaches its threshold; requests with a conjunction go straight to Tier 4. On CLINC150 the cascade is Tier 3 → Tier 4. Latencies are orders of magnitude on the test machine (Table 2). The planner, tools and memory serve the assistant; the evaluation scores the routing decision.

below 50 °C before each session. One process loads every model once; the systems run in rotated order per item and repetition after five warm-up requests each. Per item, latency is the median of 5 timed repetitions of the end-to-end wall time on a monotonic clock (Kestrel-Intents-v0: all items; CLINC150: a fixed, seeded 20% subset of 1100 items, with one timed run on the rest, so latency statistics use the subset and accuracy all items). Runs are cut into chunks of about one hour, each with its own pre-checks. Five pre-declared rules invalidate a chunk (lost AC power or boost on; background load above 10%; a frequency deviation above 10%; a temperature above 95 °C; a crash or model-hash mismatch). Every test run was valid at the first attempt. Model loading is excluded from latency. Energy was not measured, because the only counter (RAPL) is root-only on this machine.

Statistics. Paired bootstrap with 10 000 resamples (seed 0) (Efron and Tibshirani, 1993) gives 95% CIs. On Kestrel-Intents-v0 the bootstrap resamples template families, since items of one family are not independent (Field and Welsh, 2007). H1 is an intersection–union test (Berger, 1982): the lower CI bound of Δaccuracy above $-\delta$ *and* the upper CI bound of Δlatency below 0. The 13 secondary members get one p-value each on the two-sided scale (bootstrap for margins and non-0/1 statistics, exact McNemar (McNemar, 1947) for paired correctness, Wilcoxon signed-rank for latency and paired runs) and are adjusted together by Holm's method (Holm, 1979). Outcomes are *supported* (adjusted $p < 0.05$ in the hypothesized direction), *not supported* (the CI wholly on the wrong side) or *inconclusive*. We follow common guidance on significance testing and reporting in NLP (Dror et al., 2018; Dodge et al., 2019).

7 Baselines

All systems share the same models, prompt templates, grammar and thread count, and every threshold is chosen on validation; each single-model baseline gets one abstention threshold. **R0** always uses the SLM (Tier 4) and abstains below a tuned threshold on its calibrated confidence; **R0 untuned** always answers. **R1** uses the encoder head alone with a tuned threshold. **R2** is the cascade on raw confidences with one shared threshold, and **R2c** the same with calibrated confidences (used for RQ2). **R3** is the calibrated cascade with per-tier thresholds at the operating

Table 2: Routing policies on the test sets (E1; Table R1 of the protocol). Accuracy counts abstention on an in-scope item as an error. Latency is per request on a warm process, model loading excluded; "Tier 4" is the share of requests that reached the SLM. Kestrel-Intents-v0 is synthetic.

CLINC150 ($n = 5500$; latency on 1100 timed items)

System	Accuracy	Macro-F1	Coverage	Mean ms	Median ms	p95 ms	Peak RSS MiB	Tier 4
R0	0.827	0.870	0.875	863	865.0	1292	3491	100%
R0 untuned	0.773	0.850	1.000	863	865.0	1292	3491	100%
R1	0.893	0.925	0.811	6	5.4	8	278	0%
R3	0.874	0.914	0.890	146	5.6	931	3773	17%

Kestrel-Intents-v0, synthetic ($n = 252$, all timed)

System	Accuracy	Macro-F1	Coverage	Mean ms	Median ms	p95 ms	Peak RSS MiB	Tier 4
R0	1.000	1.000	1.000	689	669.5	916	3367	100%
R0 untuned	1.000	1.000	1.000	689	669.5	916	3367	100%
R1	0.702	0.860	1.000	6	5.6	9	276	0%
R3	0.968	0.989	1.000	382	521.7	917	3638	54%

point of Eq. (2). **R-rand** escalates at random with R3's per-tier escalation rates, so its expected cost matches R3's; its accuracy is an expectation over at least 1 000 seeds. For RQ5 the baselines are naive fine-tuning, online EWC (Kirkpatrick et al., 2017) and replay, with joint training as an upper bound and the stage-0 head as a lower bound. Naive, EWC and replay each get the same budget of 12 configurations; the two bounds use naive's selected learning rate and epochs.

8 Evaluation Metrics

Accuracy uses the scoring rule of Section 3; *macro-F1* averages over gold classes, counting an abstention as the OOS label; *coverage* is the share of requests answered. *Selective risk* at coverage 0.80 is the error rate among answered requests at the smallest coverage ≥ 0.80. *AURC* is the area under the risk–coverage curve on a 20-point grid. *ECE* uses 15 equal-width, right-closed bins. *Latency* is reported as the mean, median and nearest-rank p95 over per-item medians; *peak RSS* is `VmHWM` of a process that loads only one system's models. For RQ5, with $R_{i,j}$ the accuracy on stage j after training stage i of S stages, we report average final accuracy, average forgetting (Chaudhry et al., 2018) and backward transfer (Lopez-Paz and Ranzato, 2017).

9 Results

9.1 RQ1: the cascade against the single models (E1)

H1, primary (CLINC150): supported. At its validation-chosen operating point the cascade sent 17.0% of requests to the SLM (which answered 6.0% of all requests and abstained on 11.0%) and reached accuracy 0.874 against 0.827 for the tuned SLM, Δaccuracy $+0.048$ (95% CI $[+0.038, +0.058]$), with Δmean latency -717 ms (CI $[-739, -694]$); see Table 2. Both parts of the intersection–union test hold, and the result is supported. Dropping the 14 test items that also occur in other splits ($n = 5486$) leaves the accuracy CI at $[+0.038, +0.058]$. The median latency of requests that the cascade answered below Tier 4 was 5.3 ms, against 865.0 ms for the SLM. The p95 latency, 931 ms, is set by the requests that do reach Tier 4. The cascade does not save memory: it keeps the encoder loaded next to the SLM, and its peak RSS was 3773 MiB against 3491 MiB.

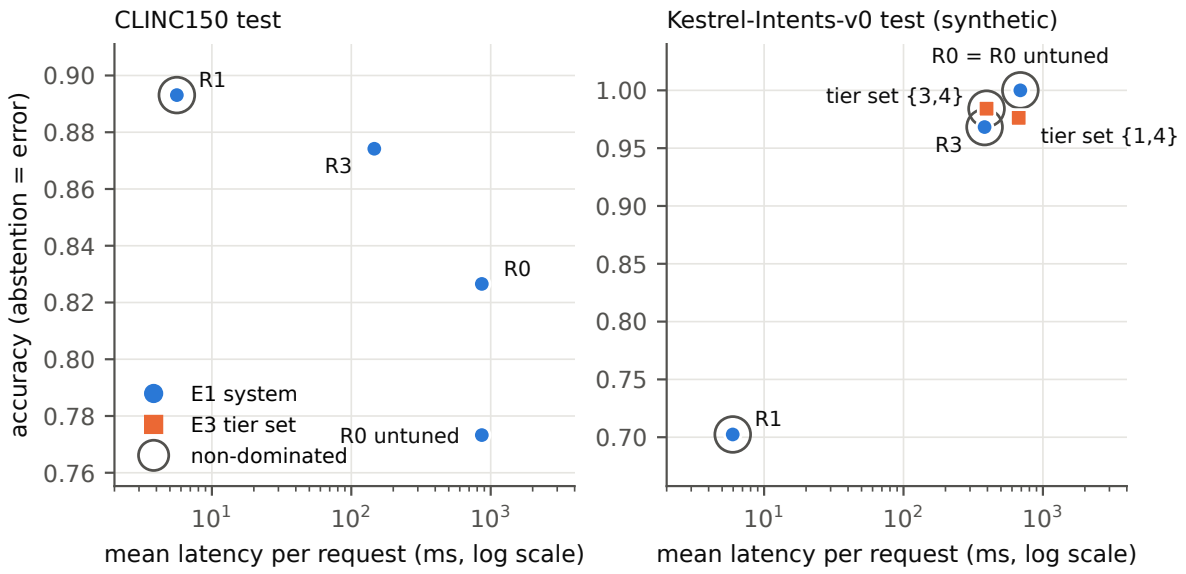


Figure 3: Accuracy against mean latency on the test sets: the E1 systems and the extra tier sets of E3. Ringed points are not dominated (no other point is both at least as accurate and at least as fast). On CLINC150 the encoder alone (R1) dominates the cascade (R3).

H1b: better than random escalation at the same cost (supported on both datasets). R-rand, which escalates as often as R3 at random, reached 0.716 on CLINC150 (R3 − R-rand: +0.159, CI [+0.151, +0.166]) and 0.912 on Kestrel-Intents-v0 (+0.056, CI [+0.021, +0.098]). The confidence signal, not just the escalation rate, is what earns the accuracy.

H1c: against the encoder alone (CLINC150 inconclusive, Kestrel-Intents-v0 supported). On CLINC150 the encoder alone was more accurate than the cascade (0.893 vs 0.874; Δ −0.019, CI [−0.026, −0.012]) and faster (mean 6 vs 146 ms), so R1 dominates R3 on the test frontier (Figure 3). The CI's lower bound lies just inside the margin (unadjusted $p = 0.045$), but not after Holm adjustment ($p = 0.27$). On Kestrel-Intents-v0 the encoder alone was far weaker (0.702) and the cascade gained +0.266 (CI [+0.154, +0.384]). Section 11 explains the CLINC150 result.

H1 on Kestrel-Intents-v0: inconclusive. Here the SLM was perfect (1.000). The cascade was faster (Δmean latency −307 ms, CI [−385, −229]) but less accurate (Δaccuracy −0.032, CI [−0.069, −0.004]). The CI crosses $-\delta$, so non-inferiority is not shown, and its upper bound is below 0.

RQ7, H7a: Hinglish (inconclusive). On Hinglish requests the cascade stayed close to the SLM (Δ −0.008, CI [−0.024, +0.000]; unadjusted $p = 0.036$, Holm $p = 0.25$); on English it lost more (Δ −0.056, CI [−0.128, +0.000]; reported, not tested). Hinglish requests reached the SLM more often (66.7% vs 40.5%), so they cost more (mean 472 ms vs 291 ms); this is descriptive, as pre-registered.

9.2 RQ2: calibration (E2)

Temperature scaling reduced the encoder head's ECE on CLINC150 from 0.0553 to 0.0083 ($T_3 = 0.666$, so the raw head was under-confident) and Tier 4's from 0.0489 to 0.0216 (Table 3). A temperature never changes a tier's predicted label, but over many classes it does change the order of confidences across requests (Kendall τ between Tier 3's raw and calibrated confidences: 0.92

Table 3: Per-tier calibration on test (E2; Table R2), on single-intent items whose gold label is a class of the Tier-3 head: in-scope items on CLINC150, and in-scope plus out-of-scope items on Kestrel-Intents-v0, whose head has an `unsupported` class. T is fitted on the calibration split; risk is at coverage 0.80 within the tier.

Data	Tier	n	Acc.	T	ECE raw	ECE cal.	Risk@0.8 raw	Risk@0.8 cal.
CLINC150	Tier 3	4500	0.959	0.666	0.0553	0.0083	0.0058	0.0058
CLINC150	Tier 4	4500	0.904	1.263	0.0489	0.0216	0.0297	0.0294
Intents-v0	Tier 3	204	0.868	0.688	0.0590	0.0557	0.0671	0.0671
Intents-v0	Tier 4	204	1.000	0.847	0.0097	0.0058	0.0000	0.0000

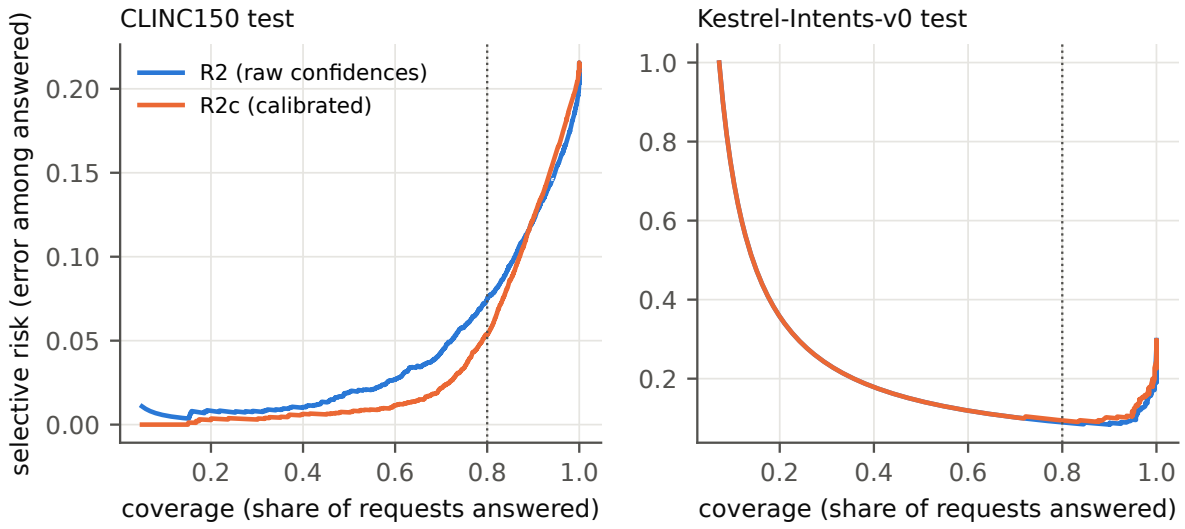


Figure 4: Risk–coverage curves of the cascade with one shared threshold on raw (R2) and calibrated (R2c) confidences, test sets. The dotted line marks coverage 0.80, where H2 is tested.

on CLINC150 test, 0.94 on Kestrel-Intents-v0). Within each tier, selective risk at coverage 0.80 still hardly moved (Table 3); calibration acts mainly by making the tiers' confidences comparable under one shared threshold.

H2 (CLINC150 supported, Kestrel-Intents-v0 inconclusive). On CLINC150, calibration lowered selective risk at coverage 0.80 from 0.0748 to 0.0539 (Δ -0.0209, CI $[-0.0263, -0.0148]$) and AURC from 0.0463 to 0.0380 (Figure 4). On Kestrel-Intents-v0 it did not replicate: risk 0.0891 vs 0.0941 (Δ $+0.0050$, CI $[-0.0007, +0.0199]$), and AURC was slightly worse with calibration (Δ $+0.0045$, CI $[+0.0004, +0.0091]$). The Tier-3 temperature fitted on 100 synthetic calibration items hardly changed that tier's ECE (0.0590 to 0.0557).

H7b: calibration by language (inconclusive). With one temperature for both languages, Tier 3's ECE was 0.0595 on English and 0.1011 on Hinglish (gap $+0.0416$, CI $[-0.0023, +0.1106]$; unadjusted $p = 0.06$).

9.3 RQ5: forgetting in the trainable head (E5)

Updating the head on each new stage alone (naive) forgot almost everything learned earlier: average forgetting 0.947 and average accuracy 0.188, no better than never updating (stage 0, 0.188). Replay of 50 stored items per earlier class removed the forgetting (-0.002; accuracy 0.918, close to joint training's 0.928). **H5 is supported**: replay minus naive forgetting -0.949

Table 4: Class-incremental updates of the Tier-3 head on CLINC150 (E5; Table R6): 10 domains in 5 stages of 2, mean ± SD over 15 runs (3 domain orders × 5 seeds), configurations selected on validation.

Method	Avg. accuracy	Avg. forgetting	BWT	Update s	Buffer MB
naive	0.188 ± 0.005	0.947 ± 0.006	-0.947	3.6	0.0
replay	0.918 ± 0.004	-0.002 ± 0.005	+0.006	7.0	9.3
ewc	0.183 ± 0.004	0.932 ± 0.007	-0.932	6.0	0.0
joint	0.928 ± 0.003	0.012 ± 0.003	-0.011	9.5	0.0
stage0	0.188 ± 0.005	0.000 ± 0.000	+0.000	0.8	0.0

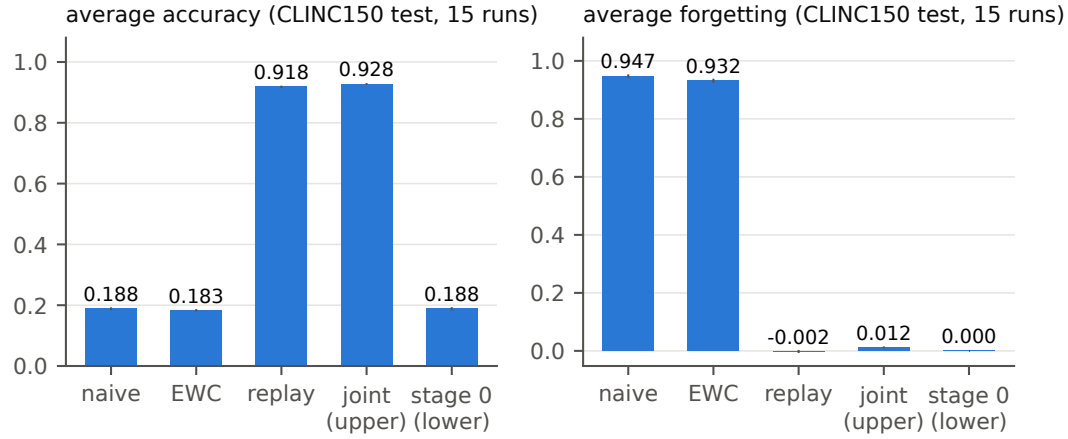


Figure 5: Average accuracy and forgetting of each update method (CLINC150 test, 15 runs; bars show the mean, error bars the SD).

(CI $[-0.953, -0.946]$; Wilcoxon over the 15 paired runs, $p = 6.1 \times 10^{-5}$). EWC's selected configuration changed forgetting by only -0.016 relative to naive's, which used a different learning rate (reported, outside the family). On the development runs, with learning rate and epochs matched, the penalty left naive's results unchanged at every penalty weight in the grid: the empirical Fisher information was small, and SGD's stability bound capped the penalty weight. The price of replay is its buffer, which stores half of every earlier class's training items (Table 4).

9.4 All pre-registered hypotheses

Table 5 lists every member: 6 of 13 are supported, none is not supported, five are inconclusive and two are pending.

10 Ablation Studies

H3: adding the encoder to rules + SLM (supported). On Kestrel-Intents-v0, adding Tier 3 to $\{1, 4\}$ kept accuracy within the margin (Δ -0.008, CI $[-0.020, +0.000]$) and lowered mean cost (Δ -289 ms, CI $[-371, -211]$), Holm $p = 0.045$ (Table 6). On CLINC150, $\{3, 4\}$ against $\{4\}$ is the H1 comparison (Δ $+0.048$, -717 ms). Tier 2 was not built, so H3's clause about it is untested.

The rules cost accuracy on test. $\{1, 4\}$ was less accurate than $\{4\}$ (0.976 vs 1.000) for a small saving (670 vs 689 ms), and $\{3, 4\}$ was more accurate than the full cascade $\{1, 3, 4\}$ (0.984 vs 0.968) at almost the same cost (394 vs 382 ms). The failure analysis (Section 11) shows why.

Table 5: The secondary family (Holm, 13 members). H4 and H6 have not run; they enter with $p = 1$ until their amendment, so every adjustment is interim: an interim "supported" is final, and an interim "inconclusive" may still change. H1 on CLINC150 is the primary and is tested alone.

Hypothesis	p	Holm p	Outcome	Source
H1 Intents-v0	1	1	inconclusive (interim)	E1 Intents-v0
H1b CLINC150	$< 10^{-4}$	$< 10^{-4}$	supported	E1 CLINC150
H1b Intents-v0	2.0×10^{-4}	0.002	supported	E1 Intents-v0
H1c CLINC150	0.045	0.27	inconclusive (interim)	E1 CLINC150
H1c Intents-v0	$< 10^{-4}$	$< 10^{-4}$	supported	E1 Intents-v0
H2 CLINC150	$< 10^{-4}$	$< 10^{-4}$	supported	E2 CLINC150
H2 Intents-v0	1	1	inconclusive (interim)	E2 Intents-v0
H3	0.006	0.045	supported	E3 Intents-v0
H4	pending	pending	pending	—
H5	6.1×10^{-5}	6.1×10^{-4}	supported	E5 CLINC150
H6	pending	pending	pending	—
H7a	0.036	0.25	inconclusive (interim)	E1 Intents-v0 hi-en
H7b	0.06	0.30	inconclusive (interim)	E2 Intents-v0

Table 6: Tier ablation on test (E3; Table R3); each set is tuned on validation by the same rule. On CLINC150 the rules cannot answer, so only $\{4\}$ and $\{3, 4\}$ apply. Cost is the sum of the consulted tiers' step times measured in E1, so its p95 can differ slightly from the end-to-end latency of the same system in Table 2.

Data	Tier set	Accuracy	Mean ms	p95 ms	Ended at Tier 1 / 3 / 4
CLINC150	{4}	0.827	863	1292	0.00 / 0.00 / 1.00
CLINC150	{3,4}	0.874	146	933	0.00 / 0.83 / 0.17
Intents-v0	{4}	1.000	689	916	0.00 / 0.00 / 1.00
Intents-v0	{1,4}	0.976	670	916	0.02 / 0.00 / 0.98
Intents-v0	{3,4}	0.984	394	912	0.00 / 0.45 / 0.55
Intents-v0	{1,3,4}	0.968	382	912	0.02 / 0.44 / 0.54

Calibration. RQ2 is the calibration ablation (Section 9). With per-tier thresholds, calibration did not make the cascade cheaper. In an exploratory check made after the test results, the validation search of Eq. (2) run on raw confidences reached the same accuracy (0.912 on CLINC150, 0.984 on Kestrel-Intents-v0) at a lower mean cost than the calibrated search that chose R3 (128 vs 140 ms; 467 vs 478 ms). A development note, repeated in the pre-registration, had stated that a temperature cannot move this frontier. It assumed that a temperature keeps the order of confidences, which holds within one request but not across requests (Section 9), and on CLINC150 Tier 4 also abstains at the operating point. Calibration's measured benefit is abstention quality under one shared threshold (H2).

11 Discussion

Each single model wins on one dataset; the cascade wins on neither. The SLM alone is best on Kestrel-Intents-v0 and the encoder alone on CLINC150 (Figure 3). The cascade sits between them: it is much better than the weaker single model on each dataset (+0.048 over the SLM on CLINC150, +0.266 over the encoder on Kestrel-Intents-v0) and much cheaper than the SLM. Its operating point was chosen to match the better single model on validation, and it did there, but not on test. The honest summary is a hedge: a deployment that cannot know in advance which model suits its traffic loses little with the cascade, but a deployment whose traffic resembles CLINC150 would do better with the encoder alone.

Why the cascade lost to the encoder on CLINC150 (exploratory). The cascade made 692 errors on CLINC150, and 474 of them (68%) were out-of-scope requests answered with an in-scope intent. The operating point accepts Tier 3 at confidence ≥ 0.601, the cheapest point that met the floor on validation, while R1 abstains below 0.782; 149 out-of-scope test requests fall between the two, and the cascade answers them. OOS requests were 11.8% of validation but 18.2% of test, so a threshold chosen for cost on validation under-weights them. Part of the loss is therefore a shift in the OOS rate, which matches the finding that confidence-based deferral can fail under distribution shift (Jitkrittum et al., 2023). Whether Tier 3 is also miscalibrated on OOS requests is open: its test ECE after scaling (0.0083) covers in-scope items only, and its temperature was fitted on in-scope calibration items, since OOS requests have no head class. Remedies to test in a new, pre-registered version: a reject band at Tier 3, an OOS detector before Tier 3, or a validation set with the deployment's OOS share.

Why the rules hurt on Kestrel-Intents-v0 (exploratory). Of the cascade's 8 errors, 6 were compound requests joined by *also* or *, now*, which the conjunction signal does not know, so a single-intent rule accepted them. All 24 validation compounds contained a signal word, so tuning could not see the gap. A wider signal is a post-hoc change and would be evaluated as such.

Practical guidance. (1) Measure the OOS share of real traffic and choose the operating point on data that matches it. (2) Report the encoder alone with a tuned threshold as a baseline: it is cheap and, on CLINC150, the system to beat. (3) A cascade saves latency, not memory: keeping a 4-bit SLM resident costs about 3491 MiB whether or not most requests use it. (4) Calibrate when one threshold compares confidences across tiers (H2); with per-tier thresholds, calibration brought no saving on validation here (Section 10).

12 Threats to Validity

Internal. Thresholds and temperatures were tuned only on calibration and validation data, and the configurations were frozen and tagged before the test sets were read. After the freeze, one clarification fixed which datasets enter the family for H1b, H1c and H2; by then E5's and Kestrel-Intents-v0's E2 results had been seen, and both datasets were kept in the family. Thermal and background effects are controlled by the pinned protocol and the five invalidation rules, and no test chunk was invalidated. Systems were interleaved per item, so slow drift affects them alike. The failure analysis was done after the results were read and is labelled exploratory.

External. One machine, one SLM and one encoder; turbo boost was off, so absolute latencies are conservative and other CPUs, GPUs or NPUs may change the ratios. Kestrel-Intents-v0 is synthetic, small, and drafted with an AI assistant, and its labels were frozen before a second annotator measured agreement. Hinglish appears only in this set.

Construct. Counting abstention as an error on in-scope items, but as correct on OOS items, rewards cautious systems on CLINC150, with its high OOS share. We report coverage and macro-F1 next to accuracy for this reason. Latency excludes model loading (tens of seconds) and is measured on a warm, single-user process. Energy was not measured.

Contamination. CLINC150 has been public since 2019 and may be in the SLM's pre-training data. This cannot be ruled out, but it would favour R0, against which the cascade was compared.

Statistical. The family's adjusted p-values are interim until H4 and H6 run. The margin δ was frozen before the mentor confirmed the smallest effect of interest. If that is set larger, H1, H1c, H3 and H7a are also reported at the larger margin as a sensitivity analysis; the primary test stays at $\delta = 0.0263$.

13 Limitations

We did not test: other hardware, other SLMs or encoders, or quantization levels (E6, H6, pending); personalization from corrections (E4, H4, pending, waiting on the review of its synthetic personas); a Tier-2 distilled model; a learned deferral gate; energy; or any public code-mixed benchmark. The per-language intervals are wide with 126 items per language. "No network" is shown for the tested scenarios only; it is not a formal privacy guarantee, nothing confines the process at kernel level, and swap, crash dumps and backups are not covered. The heads are trained in Python and not stored in git, and whether retraining gives the same bytes on another platform has not been checked.

14 Ethical Considerations

Local data handling. All inference runs on the user's machine. A privacy check (11/11 passing) shows no network crates in the dependency tree, no URL and no shell in the runtime code, zero `connect()` calls under a network namespace in the tested scenarios (with a deliberate connection as a positive control), no files opened outside the allowed roots, and a planted canary absent from telemetry and gone from the database after deletion.

Consent for learning. Kestrel asks before it stores a correction or a learned alias, and the user can list, export or delete stored items and purge expired ones. The replay buffer that prevents forgetting (RQ5) keeps an embedding of each stored request, which is derived from the user's text and should be treated as personal data; storing or training on user text carries a memorization risk (Carlini et al., 2021), and formal guarantees such as differential privacy (Abadi et al., 2016) were not applied.

Tools. Tools are allow-listed, confined to the notes folder by a path sandbox that rejects traversal and symbolic links that lead outside it, and destructive actions ask first.

Synthetic data and AI assistance. Kestrel-Intents-v0 and the draft persona and task sets are synthetic, were drafted with an AI coding assistant, and describe no real person. The code and analysis scripts were also written with that assistant under the author's review; the pre-registration, design decisions and their rationale are the author's and are logged with dates.

15 Reproducibility

The repository pins the Rust toolchain and crates (`Cargo.lock`), Python packages (`requirements.lock`), every model by repository, revision and SHA-256 (the encoder GGUF is converted locally by a pinned script), and the datasets by commit and SHA-256. Every run writes a manifest with the environment, hashes, checks and a validity flag. One command (`python experiments/repro.py --rerun-e5`) clones the committed state into a scratch directory, regenerates every analysis from the raw logs and compares the outputs. At commit `74def15`, 11 of 11 outputs were byte-identical, and the E5 rerun gave equal numbers. This paper's numbers, tables and figures are generated from the same logs (`experiments/analysis/paper_assets.py`). A

reproduction on a second machine is planned and has not been done. Licences: CLINC150 CC BY 3.0, Gemma-4 Apache-2.0, multilingual-e5-small MIT, Kestrel's code MIT or Apache-2.0.

16 Conclusion

On one laptop CPU and under a frozen protocol, the calibrated cascade (encoder then SLM on CLINC150, with keyword rules in front on the synthetic set) was as accurate as always using the SLM, and faster, on CLINC150 (Δaccuracy $+0.048$, Δmean latency -717 ms per request; RQ1, H1 supported), and its confidence signal beat random escalation at equal cost on both datasets (H1b). Calibration lowered selective risk on CLINC150 but not on the synthetic set (RQ2). Adding the encoder to rules and SLM changed mean cost by -289 ms per request with accuracy within the margin (RQ3), and replay removed the head's forgetting (RQ5). What was *not* supported matters as much. The cascade did not beat the encoder alone on CLINC150, where the encoder dominated it, or the SLM alone on the synthetic set. The cascade's advantage over the single models therefore depends on traffic, notably on the out-of-scope share. Its Hinglish behaviour (RQ7) remains inconclusive, and personalization and quantization are still to run.

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